חשבון אינפי 1 - 20151

דף תרגילים מס' 10.

אינטגרל לא מסוים:
טבלת האינטגרלים, הצבה ואינטגרציה בחלקים,
אינטגרציה של פונקציות רציונאליות ,אי-רציונאליות וטריגונומטריות.

פתרונות ותשובות

1. (א) $\int\left(2x+3e^x-x\sqrt{x}\right)dx=x^2+3e^x-\frac{2}{5}x^2\sqrt{x}+C$,

(ב) $\int\left(5\cos x+3^x+\frac{1}{2\sqrt[3]{x}}\right)dx=5\sin x+\frac{3^x}{\ln 3}+\frac{3}{4}\sqrt[3]{x^2}+C$,

(ג) $\int\frac{(18+2x^2)\sin x+5}{x^2+9}dx=\int\left(2\sin x+\frac{5}{x^2+9}\right)dx=-2\cos x+\frac{5}{3}\arctan\frac{x}{3}+C$,

(ד) $\int\frac{\left(\sqrt{x}+2\right)^2}{\sqrt{x}}dx=\int\left(\sqrt{x}+4+\frac{4}{\sqrt{x}}\right)dx=\frac{2}{3}\sqrt{x^3}+4x+8\sqrt{x}+C$,

(ה) $\int\frac{3-\cos^2 x}{\cos^2 x}dx=\int\left(\frac{3}{\cos^2 x}-1\right)dx=3\tan x-x+C$,

(ו) $\int\sqrt[3]{5-6x}dx=-\frac{1}{8}\sqrt[3]{(5-6x)^4}+C$,

(ז) $\int\frac{2x^3+8x-5}{4+x^2}dx=\int\left(2x-\frac{5}{4+x^2}\right)dx=x^2-\frac{5}{2}\arctan\frac{x}{2}+C$,

(ח) $\int 3^{2-3x}dx=-\frac{3^{2-3x}}{3\ln 3}+C$,

(ט) $\int\frac{dx}{1+2x^2}=\frac{1}{2}\int\frac{dx}{\frac{1}{2}+x^2}=\frac{1}{\sqrt{2}}\arctan\left(x\sqrt{2}\right)+C$,

(י) $\int\left(\frac{2}{9+x^2}+\frac{5}{x^2}\right)dx=\frac{2}{3}\arctan\frac{x}{3}-\frac{5}{x}+C$,

(יא) $\int\frac{dx}{\sqrt{36-3x^2}}=\frac{1}{\sqrt{3}}\int\frac{dx}{\sqrt{12-x^2}}=\frac{1}{\sqrt{3}}\arcsin\frac{x}{\sqrt{12}}+C$,

2. (א) $\int e^{-2x+1}dx=-\frac{1}{2}e^{-2x+1}+C$,

(ב) $\int\frac{dx}{1-3x}=-\frac{1}{3}\ln|1-3x|+C$,

$$,\int \frac{\tan^2 x + 3x\sin^2 x}{\sin^2 x}dx = \int\left(\frac{1}{\cos^2 x} + 3x\right)dx = \tan x + \frac{3}{2}x^2 + C \quad (ג)$$

$$,\int \frac{dx}{9-4x^2} = \frac{1}{4}\int \frac{dx}{\frac{9}{4}-x^2} = \frac{1}{4}\frac{1}{3}\ln\left|\frac{\frac{3}{2}+x}{\frac{3}{2}-x}\right| + C = \frac{1}{12}\ln\left|\frac{3+2x}{3-2x}\right| + C \quad (ד)$$

$$,\int \frac{1+x+3x^2}{\sqrt{x}}dx = \int\left(\frac{1}{\sqrt{x}} + \sqrt{x} + 3x\sqrt{x}\right)dx = 2\sqrt{x} + \frac{2}{3}x\sqrt{x} + \frac{6}{5}x^2\sqrt{x} + C \quad (ה)$$

$$,\int\left(e^{5x} - \frac{2x^2+5}{1+x^2}\right)dx = \int\left(e^{5x} - 2 - \frac{3}{1+x^2}\right)dx = \frac{1}{5}e^{5x} - 2x - 3\arctan x + C \quad (ו)$$

$$,\int \frac{dx}{(5-3x)^2} = \frac{1}{3(5-3x)} + C \quad (ז)$$

$$,\int \frac{dx}{3-16x^2} = \frac{1}{16}\int \frac{dx}{\frac{3}{16}-x^2} = \frac{1}{8\sqrt{3}}\ln\left|\frac{\frac{\sqrt{3}}{4}+x}{\frac{\sqrt{3}}{4}-x}\right| + C = \frac{1}{8\sqrt{3}}\ln\left|\frac{\sqrt{3}+4x}{\sqrt{3}-4x}\right| + C \quad (ח)$$

$$,\int \frac{dx}{\sqrt{4-3x^2}} = \frac{1}{\sqrt{3}}\int \frac{dx}{\sqrt{\frac{4}{3}-x^2}} = \frac{1}{\sqrt{3}}\arcsin\frac{x\sqrt{3}}{2} + C \quad (ט)$$

$$,\int \frac{dx}{\sqrt{1-(4x+2)^2}} = \frac{1}{4}\arcsin(4x+2) + C \quad (י)$$

$$.\int \frac{dx}{1-5x^2} = \frac{1}{5}\int \frac{dx}{\frac{1}{5}-x^2} = \frac{\sqrt{5}}{2}\ln\left|\frac{\frac{1}{\sqrt{5}}+x}{\frac{1}{\sqrt{5}}-x}\right| + C = \frac{\sqrt{5}}{2}\ln\left|\frac{1+x\sqrt{5}}{1-x\sqrt{5}}\right| + C \quad (יא)$$

3.

$$,\int \frac{4x^3-1}{x^4-x+5}dx = \left|\begin{matrix} t = x^4-x+5 \\ dt = (4x^3-1)dx \end{matrix}\right| = \int \frac{dt}{t} = \ln|t| + C = \ln\left|x^4-x+5\right| + C \quad (א)$$

$$,\int \frac{\sqrt[4]{\ln x}}{x}dx = \left|\begin{matrix} t = \ln x \\ dt = \frac{dx}{x} \end{matrix}\right| = \int \sqrt[4]{t}\,dt = \frac{4}{5}\sqrt[4]{t^5} + C = \frac{4}{5}\sqrt[4]{\ln^5 x} + C \quad (ב)$$

$$,\int \frac{3xdx}{(x^2+4)^3} = \left|\begin{matrix} t = x^2+4 \\ dt = 2xdx \end{matrix}\right| = \frac{3}{2}\int \frac{dt}{t^3} = -\frac{3}{4t^2} + C = -\frac{3}{4(x^2+4)^2} + C \quad (ג)$$

$$,\int \frac{e^{\tan x}}{\cos^2 x}dx = \left|\begin{matrix} t = \tan x \\ dt = \frac{dx}{\cos^2 x} \end{matrix}\right| = \int e^t dt = e^t + C = e^{\tan x} + C \quad (ד)$$

$$\int \frac{\cos xdx}{\sqrt{\sin^2 x+3}} = \left|\begin{matrix} t = \sin x \\ dt = \cos xdx \end{matrix}\right| = \int \frac{dt}{\sqrt{t^2+3}} = \ln\left|t+\sqrt{t^2+3}\right| + C = \quad (ה)$$

$$= \ln\left|\sin x + \sqrt{\sin^2 x + 3}\right| + C$$

ו) $\int \frac{\sqrt{\arctan x}}{1+x^2}dx = \left|\begin{matrix} t = \arctan x \\ dt = \frac{dx}{1+x^2}\end{matrix}\right| = \int \sqrt{t}dt = \frac{2}{3}\sqrt{t^3} + C = \frac{2}{3}\sqrt{\arctan^3 x} + C$,

ז) $\int \sqrt[3]{2+7\sin x}\cos xdx = \left|\begin{matrix} t = 2+7\sin x \\ dt = 7\cos xdx\end{matrix}\right| = \frac{1}{7}\int \sqrt[3]{t}dt = \frac{3}{28}\sqrt[3]{t^4} + C = \frac{3}{28}\sqrt[3]{(2+7\sin x)^4} + C$,

ח) $\int \frac{1}{x^2}\cos\frac{1}{x}dx = \left|\begin{matrix} t = \frac{1}{x} \\ dt = -\frac{dx}{x^2}\end{matrix}\right| = -\int \cos tdt = -\sin t + C = -\sin\frac{1}{x} + C$,

ט) $\int \frac{\ln x + 2}{2x}dx = \left|\begin{matrix} t = \ln x + 2 \\ dt = \frac{dx}{x}\end{matrix}\right| = \frac{1}{2}\int tdt = \frac{1}{4}t^2 + C = \frac{1}{4}(\ln x + 2)^2 + C$,

י) $\int \frac{\cos 5x}{\sin^2 5x + 1}dx = \left|\begin{matrix} t = \sin 5x \\ dt = 5\cos 5xdx\end{matrix}\right| = \frac{1}{5}\int \frac{dt}{t^2+1} = \frac{1}{5}\arctan t + C = \frac{1}{5}\arctan(\sin 5x) + C$,

יא) $\int \frac{dx}{\sin^2 x\sqrt{3-\cot x}} = \left|\begin{matrix} t = 3-\cot x \\ dt = \frac{dx}{\sin^2 x}\end{matrix}\right| = \int \frac{dt}{\sqrt{t}} = 2\sqrt{t} + C = 2\sqrt{3-\cot x} + C$.

.4 א) $\int e^{-x}(x^2+5)dx = \left|\begin{matrix} u = x^2+5 & du = 2xdx \\ dv = e^{-x}dx & v = -e^{-x}\end{matrix}\right| = -(x^2+5)e^{-x} + 2\int xe^{-x}dx =$

$$= \left|\begin{matrix} u_1 = x & du_1 = dx \\ dv_1 = e^{-x}dx & v_1 = -e^{-x}\end{matrix}\right| = -(x^2+5)e^{-x} + 2\left(-xe^{-x} + \int e^{-x}dx\right) =$$

$$= -(x^2+5)e^{-x} - 2xe^{-x} - 2e^{-x} + C$$

ב) $\int \arcsin xdx = \left|\begin{matrix} u = \arcsin x & du = \frac{dx}{\sqrt{1-x^2}} \\ dv = dx & v = x\end{matrix}\right| = x\arcsin x - \int \frac{xdx}{\sqrt{1-x^2}} =$

$$= x\arcsin x + \sqrt{1-x^2} + C$$

$$\int \frac{xdx}{\sqrt{1-x^2}} = \left|\begin{matrix} t = 1-x^2 \\ dt = -2xdx\end{matrix}\right| = -\frac{1}{2}\int \frac{dt}{\sqrt{t}} = -\sqrt{t} + C = -\sqrt{1-x^2} + C$$

ג) $\int x\sin 3xdx = \left|\begin{matrix} u = x & du = dx \\ dv = \sin 3xdx & v = -\frac{1}{3}\cos 3x\end{matrix}\right| = -\frac{1}{3}x\cos 3x + \frac{1}{3}\int \cos 3xdx =$

$$= -\frac{1}{3}x\cos 3x + \frac{1}{9}\sin 3x + C$$

ד) $I = \int e^x\sin 3xdx = \left|\begin{matrix} u = e^x & du = e^xdx \\ dv = \sin 3xdx & v = -\frac{1}{3}\cos 3x\end{matrix}\right| = -\frac{1}{3}e^x\cos 3x + \frac{1}{3}\int e^x\cos 3xdx =$

$$=\left|\begin{matrix} u_1=e^x & du_1=e^x dx \\ dv_1=\cos 3x dx & v=\frac{1}{3}\sin 3x \end{matrix}\right|=-\frac{1}{3}e^x\cos 3x+\frac{1}{3}\left(\frac{1}{3}e^x\sin 3x-\frac{1}{3}\int e^x\sin 3x dx\right)=$$

$$=-\frac{1}{3}e^x\cos 3x+\frac{1}{9}e^x\sin 3x-\frac{1}{9}I=\frac{9}{10}e^x\left(-\frac{1}{3}\cos 3x+\frac{1}{9}\sin 3x\right)+C$$

$$\int x\cdot 10^x dx=\left|\begin{matrix} u=x & du=dx \\ dv=10^x dx & v=\frac{10^x}{\ln 10} \end{matrix}\right|=\frac{x10^x}{\ln 10}-\frac{1}{\ln 10}\int 10^x dx=\frac{x10^x}{\ln 10}-\frac{10^x}{\ln^2 10}+C$$ (ה)

$$\int (x-2)\ln 2x dx=\left|\begin{matrix} u=\ln 2x & du=\frac{dx}{x} \\ dv=(x-2)dx & v=\frac{x^2}{2}-2x \end{matrix}\right|=\left(\frac{x^2}{2}-2x\right)\ln 2x-\int\left(\frac{x}{2}-2\right)dx=$$ (ו)

$$=\left(\frac{x^2}{2}-2x\right)\ln 2x-\left(\frac{x^2}{4}-2x\right)+C$$

$$\int \arccos 2x dx=\left|\begin{matrix} u=\arccos 2x & du=-\frac{2dx}{\sqrt{1-4x^2}} \\ dv=dx & v=x \end{matrix}\right|=x\arccos x+2\int\frac{xdx}{\sqrt{1-4x^2}}=$$ (ז)

$$=x\arccos x-\frac{1}{2}\sqrt{1-4x^2}+C$$

$$\int\frac{xdx}{\sqrt{1-4x^2}}=\left|\begin{matrix} t=1-4x^2 \\ dt=-8xdx \end{matrix}\right|=-\frac{1}{8}\int\frac{dt}{\sqrt{t}}=-\frac{1}{4}\sqrt{t}+C=-\frac{1}{4}\sqrt{1-4x^2}+C$$

$$\int \ln(x+2)dx=\left|\begin{matrix} u=\ln(x+2) & du=\frac{dx}{x+2} \\ dv=dx & v=x \end{matrix}\right|=x\ln(x+2)-\int\frac{xdx}{x+2}=$$ (ח)

$$=x\ln(x+2)-\int\left(1-\frac{2}{x+2}\right)dx=x\ln(x+2)-x+2\ln(x+2)+C$$

$$\int x^2\ln x dx=\left|\begin{matrix} u=\ln x & du=\frac{dx}{x} \\ dv=x^2dx & v=\frac{x^3}{3} \end{matrix}\right|=\frac{x^3}{3}\ln x-\frac{1}{3}\int x^2dx=\frac{x^3}{3}\ln x-\frac{1}{9}x^3+C$$ (ט)

$$I=\int\cos(\ln x)dx=\left|\begin{matrix} u=\cos(\ln x) & du=-\frac{\sin(\ln x)dx}{x} \\ dv=dx & v=x \end{matrix}\right|=x\cos(\ln x)+\int\sin(\ln x)dx=$$ (י)

$$=\left|\begin{matrix} u_1=\sin(\ln x) & du_1=\frac{\cos(\ln x)dx}{x} \\ dv_1=dx & v_1=x \end{matrix}\right|=x\cos(\ln x)+x\sin(\ln x)-\int\cos(\ln x)dx$$

$$I=\frac{x}{2}\big(\cos(\ln x)+\sin(\ln x)\big)+C$$

(יא)
$$\int \ln(x^2+1)dx = \left| \begin{matrix} u=\ln(x^2+1) & du=\dfrac{2xdx}{x^2+1} \\ dv=dx & v=x \end{matrix} \right| = x\ln(x^2+1) - 2\int \frac{x^2dx}{x^2+1} =$$
$$= x\ln(x^2+1) - 2\int \frac{x^2+1-1}{x^2+1}dx = x\ln(x^2+1) - 2x + 2\arctan x + C$$

(יב)
$$\int x^2\cos^2 xdx = \frac{1}{2}\int x^2(1+\cos 2x)dx = \frac{1}{6}x^3 + \frac{1}{2}\int x^2\cos 2xdx =$$
$$= \frac{x^3}{6} + \frac{x^2\sin 2x}{4} + \frac{x\cos 2x}{4} - \frac{\sin 2x}{8} + C$$
$$\int x^2\cos 2xdx = \left| \begin{matrix} u=x^2 & du=2xdx \\ dv=\cos 2xdx & v=\dfrac{\sin 2x}{2} \end{matrix} \right| = \frac{x^2\sin 2x}{2} - \int x\sin 2xdx =$$
$$= \left| \begin{matrix} u_1=x & du_1=dx \\ dv_1=\sin 2xdx & v_1=-\dfrac{\cos 2x}{2} \end{matrix} \right| = \frac{x^2\sin 2x}{2} - \left(-\frac{x\cos 2x}{2} + \frac{1}{2}\int \cos 2xdx \right) =$$
$$= \frac{x^2\sin 2x}{2} + \frac{x\cos 2x}{2} - \frac{\sin 2x}{4} + C$$

.5 (א) $\int \frac{10dx}{15+4x} = \frac{5}{2}\ln|15+4x| + C$,

(ב) $\int \frac{dx}{4-7x} = -\frac{1}{7}\ln|4-7x| + C$,

(ג) $\int \frac{dx}{(x+15)^{20}} = -\frac{1}{19(x+15)^{19}} + C$,

(ד) $\int \frac{dx}{(5-2x)^8} = \frac{1}{14(5-2x)^7} + C$,

(ה)
$$\int \frac{xdx}{x^2-7x+20} = \left| \begin{matrix} x^2-7x+20 = \left(x-\dfrac{7}{2}\right)^2 + \dfrac{31}{4} \\ t=x-\dfrac{7}{2},\ \ x=t+\dfrac{7}{2}, dt=dx \end{matrix} \right| = \int \frac{t+\frac{7}{2}}{t^2+\frac{31}{4}}dt = \int \frac{tdt}{t^2+\frac{31}{4}} +$$
$$+ \frac{7}{2}\int \frac{dt}{t^2+\frac{31}{4}} = \frac{1}{2}\ln\left(t^2+\frac{31}{4}\right) + \frac{7}{\sqrt{31}}\arctan\frac{2t}{\sqrt{31}} + C =$$
$$= \frac{1}{2}\ln(x^2-7x+20) + \frac{7}{\sqrt{31}}\arctan\frac{2x-7}{\sqrt{31}} + C$$

(ו)
$$\int \frac{x}{2x^2+2x+5}dx = \frac{1}{2}\int \frac{x}{x^2+x+5/2}dx = \left| \begin{matrix} t=x+\dfrac{1}{2}, dx=dt \\ x=t-\dfrac{1}{2} \end{matrix} \right| = \frac{1}{2}\int \frac{t-\frac{1}{2}}{t^2+\frac{9}{4}}dt =$$

$$=\frac{1}{2}\int\frac{tdt}{t^2+\frac{9}{4}}-\frac{1}{4}\int\frac{dt}{t^2+\frac{9}{4}}=\frac{1}{4}\ln\left(t^2+\frac{9}{4}\right)-\frac{1}{4}\frac{2}{3}\arctan\frac{2t}{3}+C=$$

$$=\frac{1}{4}\ln\left(x^2+x+\frac{5}{2}\right)-\frac{1}{6}\arctan\frac{2x+1}{3}+C$$

(ז) $\int\frac{x+1}{5x^2+2x+1}dx=\frac{1}{10}\ln\left(5x^2+2x+1\right)+\frac{1}{5}\arctan\frac{5x+1}{2}+C$,

(ח) $$\int\frac{2x+1}{x^2-4x+5}dx=\left|\begin{array}{l}x^2-4x+5=(x-2)^2+1\\ t=x-2,x=t+2,dx=dt\end{array}\right|=\int\frac{2t+5}{t^2+1}dt=\int\frac{2tdt}{t^2+1}+5\int\frac{dt}{t^2+1}=$$

$$=\ln(t^2+1)+5\arctan t+C=\ln(x^2-4x+5)+5\arctan(x-2)+C$$

(ט) $$\int\frac{dx}{2x^2-2x+3}=\frac{1}{2}\int\frac{dx}{x^2-x+\frac{3}{2}}=\frac{1}{2}\int\frac{dx}{\left(x-\frac{1}{2}\right)^2+\frac{5}{4}}=\frac{1}{2}\frac{2}{\sqrt{5}}\arctan\frac{2\left(x-\frac{1}{2}\right)}{\sqrt{5}}+C=$$

$$=\frac{1}{\sqrt{5}}\arctan\frac{2x-1}{\sqrt{5}}+C$$

.6 (א) $\int\frac{3x^3+5x^2-25x-1}{(x+2)(x-1)^2}dx=\int\left(3+\frac{5x^2-16x-7}{(x+2)(x-1)^2}\right)dx$,

$$\frac{5x^2-16x-7}{(x+2)(x-1)^2}=\frac{A}{x+2}+\frac{B}{(x-1)^2}+\frac{C}{x-1}=\frac{A(x-1)^2+B(x+2)+C(x-1)(x+2)}{(x+2)(x-1)^2}$$

$$x=1,3B=-18,B=-6,x=-2,9A=45,A=5,x=0,A+2B-2C=-7,C=0$$

$$\int\frac{3x^3+5x^2-25x-1}{(x+2)(x-1)^2}dx=\int\left(3+\frac{5}{x+2}-\frac{6}{(x-1)^2}\right)dx=3x+5\ln|x+2|+\frac{6}{x-1}+C$$

(ב) $$\frac{x^2-3x+17}{(x+5)(x^2+2x+4)}=\frac{A}{x+5}+\frac{Bx+C}{x^2+2x+4}=\frac{A(x^2+2x+4)+(Bx+C)(x+5)}{(x+5)(x^2+2x+4)},$$

$$\begin{cases}A+B=1,\\ 2A+5B+C=-3,\\ 4A+5C=17,\end{cases}\begin{cases}A=3,\\ B=-2,\\ C=1,\end{cases}$$

$$\int\frac{x^2-3x+17}{(x+5)(x^2+2x+4)}dx=\int\left(\frac{3}{x+5}-\frac{2x-1}{x^2+2x+4}\right)dx=$$

$$=3\ln|x+5|-\ln(x^2+2x+4)+\sqrt{3}\arctan\frac{x+1}{\sqrt{3}}+C$$

(ג) $\int\frac{x^2+3}{(x+1)^2(x-1)}dx$,

$$\frac{x^2+3}{(x+1)^2(x-1)}=\frac{A}{(x+1)^2}+\frac{B}{x+1}+\frac{C}{x-1}=\frac{A(x-1)+B(x-1)(x+1)+C(x+1)^2}{(x+1)^2(x-1)}$$

$$x=1, 4C=4, C=1, x=-1, -2A=4, A=-2, x=0, -A-B+C=3, B=0$$

$$\int \frac{x^2+3}{(x+1)^2(x-1)}dx=\int\left(-\frac{2}{(x+1)^2}+\frac{1}{x-1}\right)dx=\frac{2}{x+1}+\ln|x-1|+C$$

(ד) $\int \frac{x^3+2x+5}{x^2-1}dx=\int\left(x+\frac{3x+5}{x^2-1}\right)dx$,

$$\frac{3x+5}{(x-1)(x+1)}=\frac{A}{x-1}+\frac{B}{x+1}=\frac{A(x+1)+B(x-1)}{(x-1)(x+1)}, x=1, A=4, x=-1, B=-1$$

$$\int \frac{x^3+2x+5}{x^2-1}dx=\int\left(x+\frac{4}{x-1}-\frac{1}{x+1}\right)dx=\frac{x^2}{2}+4\ln|x-1|-\ln|x+1|+C$$

(ה) $$\int \frac{x^3+3x^2+5x+7}{x^2+2}dx=\int\left(x+3+\frac{3x+1}{x^2+2}\right)dx=\int\left(x+3+\frac{3x}{x^2+2}+\frac{1}{x^2+2}\right)dx=$$

$$=\frac{x^2}{2}+3x+\frac{3}{2}\ln\left(x^2+2\right)+\frac{1}{\sqrt{2}}\arctan\frac{x}{\sqrt{2}}+C$$

(ו) $\int \frac{x^2+2x+6}{(x-1)(x-2)(x-4)}dx$,

$$\frac{x^2+2x+6}{(x-1)(x-2)(x-4)}=\frac{A}{x-1}+\frac{B}{x-2}+\frac{C}{x-4}=\frac{A(x-2)(x-4)+B(x-1)(x-4)+C(x-1)(x-2)}{(x-1)(x-2)(x-4)}$$

$$A(x-2)(x-4)+B(x-1)(x-4)+C(x-1)(x-2)=x^2+2x+6$$

$$\begin{matrix} x=1 \\ x=2 \\ x=4 \end{matrix}\begin{cases} 3A=9, \\ -2B=14, \\ 6C=30, \end{cases} A=3, B=-7, C=5,$$

$$\int \frac{x^2+2x+6}{(x-1)(x-2)(x-4)}dx=\int\left(\frac{3}{x-1}-\frac{7}{x-2}+\frac{5}{x-4}\right)dx=3\ln|x-1|-7\ln|x-2|+5\ln|x-4|+C$$

(ז) $\int \frac{2x^2+x+3}{(x-1)(x^2+2x+3)}dx$,

$$\frac{2x^2+x+3}{(x-1)(x^2+2x+3)}=\frac{A}{x-1}+\frac{Bx+C}{x^2+2x+3}=\frac{A(x^2+2x+3)+(Bx+C)(x-1)}{(x-1)(x^2+2x+3)}$$

$$\begin{cases} A+B=2, \\ 2A-B+C=1, \\ 3A-C=3, \end{cases}\begin{cases} A=1, \\ B=1, \\ C=0, \end{cases}$$

$$\int \frac{2x^2+x+3}{(x-1)(x^2+2x+3)}dx=\int\left(\frac{1}{x-1}+\frac{x}{x^2+2x+3}\right)dx=$$

$$=\int\left(\frac{1}{x-1}+\frac{x+1}{(x+1)^2+2}-\frac{1}{(x+1)^2+2}\right)dx=\ln|x-1|+\frac{1}{2}\ln(x^2+2x+3)-\frac{1}{\sqrt{2}}\arctan\frac{x+1}{\sqrt{2}}+C$$

(ח) $\int \frac{dx}{x^5-x^2}$,

$$\frac{1}{x^5-x^2}=\frac{1}{x^2(x^3-1)}=\frac{1}{x^2(x-1)(x^2+x+1)}=\frac{A}{x^2}+\frac{B}{x}+\frac{C}{x-1}+\frac{Dx+E}{x^2+x+1}$$

$$A=-1,B=0,C=\frac{1}{3},D=-\frac{1}{3},E=\frac{1}{3}$$

$$\int\frac{dx}{x^5-x^2}=\int\left(-\frac{1}{x^2}+\frac{1}{3}\frac{1}{x-1}-\frac{1}{3}\frac{x-1}{x^2+x+1}\right)dx=\frac{1}{x}+\frac{1}{6}\ln\frac{(x-1)^2}{x^2+x+1}-\frac{2}{3\sqrt{3}}\arctan\frac{2x+1}{\sqrt{3}}+C$$

.7 (א)

$$\int\frac{\cos xdx}{1+\cos x}=\left|t=\tan\frac{x}{2}\right|=\int\frac{\frac{1-t^2}{1+t^2}\frac{2dt}{1+t^2}}{1+\frac{1-t^2}{1+t^2}}=\int\frac{1-t^2}{1+t^2}dt=\int\left(-1+\frac{2}{1+t^2}\right)dt=$$

$$=-t+2\arctan t+C=-\tan\frac{x}{2}+x+C$$

(ב)

$$\int\frac{dx}{4\sin x+3\cos x+5}=\left|t=\tan\frac{x}{2}\right|=\int\frac{\frac{2dt}{1+t^2}}{4\frac{2t}{1+t^2}+3\frac{1-t^2}{1+t^2}+5}=\int\frac{dt}{t^2+4t+4}=-\frac{1}{t+2}+C=,$$

$$=-\frac{1}{\tan\frac{x}{2}+2}+C$$

(ג)

$$\int\frac{dx}{5\sin x+3\cos x+3}=\int\frac{\frac{2dt}{1+t^2}}{5\frac{2t}{1+t^2}+3\frac{1-t^2}{1+t^2}+3}=\int\frac{dt}{5t+3}=\frac{1}{5}\ln|5t+3|+C=,$$

$$=\frac{1}{5}\ln\left|5\tan\frac{x}{2}+3\right|+C$$

(ד)

$$\int\sin^5 xdx=\int(1-\cos^2 x)^2\sin xdx=\left|\begin{matrix}t=\cos x\\ dt=-\sin x\end{matrix}\right|=-\int(1-t^2)^2dt=-t+\frac{2}{3}t^3-\frac{t^5}{5}+C=$$

$$=-\cos x+\frac{2\cos^3 x}{3}-\frac{\cos^5 x}{5}+C$$

(ה)

$$\int\sqrt[3]{\sin^2 x}\cos^3 xdx=\left|\begin{matrix}t=\sin x\\ dt=\cos xdx\end{matrix}\right|=\int\sqrt[3]{t^2}(1-t^2)dt=\frac{3}{5}\sqrt[3]{t^5}-\frac{3}{8}\sqrt[3]{t^8}+C=\frac{3}{5}\sqrt[3]{\sin^5 x}+\frac{3}{8}\sqrt[3]{\sin^8 x}+C$$

(ו)

$$\int\sin^4 2xdx=\frac{1}{4}\int(1-\cos 4x)^2dx=\frac{1}{4}\int(1-2\cos 4x+\cos^2 4x)dx=,$$

$$=\frac{1}{4}\int\left(1-2\cos 4x+\frac{1+\cos 8x}{2}\right)dx=\frac{1}{8}\int(3-4\cos 4x+\cos 8x)=\frac{3}{8}x-\frac{\sin 4x}{8}+\frac{\sin 8x}{64}+C$$

(ז)

$$\int\sin^2 x\cos^2 xdx=\frac{1}{4}\int\sin^2 2xdx=\frac{1}{8}\int(1-\cos 4x)dx=\frac{1}{8}x-\frac{\sin 4x}{32}+C,$$

$$\int \cos 3x \cos 2x dx = \frac{1}{2}\int (\cos 5x + \cos x) dx = \frac{1}{10}\sin 5x + \frac{1}{2}\sin x + C$$ (ח)

$$.\int \sin 5x \cos x dx = \frac{1}{2}\int (\sin 6x + \sin 4x) dx = -\frac{\cos 6x}{12} - \frac{\cos 4x}{8} + C$$ (ט)

.8 (א)

$$\int \frac{x+1}{\sqrt[3]{3x+1}} dx = \begin{vmatrix} 3x+1 = t^3 \\ 3dx = 3t^2 dt \end{vmatrix} = \frac{1}{3}\int \frac{t^3 - 1}{t} t^2 dt = \frac{1}{3}\int (t^4 - t) dt = \frac{1}{15}t^5 - \frac{1}{6}t^2 + C =$$

$$= \frac{1}{15}\sqrt[3]{(3x+1)^5} - \frac{1}{6}\sqrt[3]{(3x+1)^2} + C$$

$$,\int \frac{dx}{\sqrt{1-2x} - \sqrt[4]{1-2x}} = -\sqrt{1-2x} - 2\sqrt[4]{1-2x} - 2\ln\left|\sqrt[4]{1-2x} - 1\right| + C$$ (ב)

$$,\int \frac{\sqrt[6]{x}}{1+\sqrt[3]{x}} dx = \frac{6}{5}\sqrt[6]{x^5} - 2\sqrt{x} + 6\sqrt[6]{x} - 6\arctan\sqrt[6]{x} + C$$ (ג)

$$\int \frac{7x+16}{\sqrt[3]{(7x+8)^2} + 2\sqrt[3]{7x+8}} dx = \begin{vmatrix} 7x+8 = t^3 \\ 7dx = 3t^2 dt \end{vmatrix} = \frac{3}{7}\int \frac{t^3+8}{t^2+2t} t^2 dt = \frac{3}{7}\int (t^3 - 2t^2 + 4t) dt =$$ (ד)

$$= \frac{3}{28}t^4 - \frac{2}{7}t^3 + \frac{6}{7}t^2 + C = \frac{3}{28}\sqrt[3]{(7x+8)^4} - \frac{2}{7}(7x+8) + \frac{6}{7}\sqrt[3]{(7x+8)^2} + C$$